Mathematical Analysis I, 2024/25 First semester

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The course follows a similar way as "Mathematical Analysis 1" by Claudio Canuto, Anita Tabacco, Pearson, but it is not necessary to buy it. Also "Calculus" Vol. I by Tom M. Apostol, Wiley, will be useful.

From Monday to Thursday we have lectures, and on Friday we usually do exercises.

- Lecture notes:

  http://www.mat.uniroma2.it/~tanimoto/teaching/2024MA1/2024MA1.pdf

- Exercises:

  http://www.mat.uniroma2.it/~tanimoto/teaching/2024MA1/2024MA1ex.pdf

- Office hours: Tuesday 14:00–15:00 online, or send me a message on Teams for appointment
- Tutorial: Starting in Mid October
- Self-assessment test

Some tips

- Writing math.
  - LaTeX. You can try it here, and you can install the full set afterwards. You need to learn some commands, but once you know it it's very powerful. All my lecture notes and slides are written in LaTeX
  - Word processor (MS Word, Apple Pages, Open Office, Libre Office (Insert → Objects → Formula)...).
- Doing quick computations.
  - Wolfram Math Alpha You can just type some formulas in and it shows the result.
  - Programming languages. Python (I used it to make the graph of the SIR model), Java, C,$\cdots$

Sep 23. Overview of the course, prerequisites

What is analysis and why study it

In a real-world science, it is crucial to study **quantitative aspects** of the subject. When a quantity **changes** by time, one can study its change in a short time ($\Rightarrow$ differentiation) and then sum it up ($\Rightarrow$ integration). Another important problem is optimization: maximizing benefit or minimizing cost.

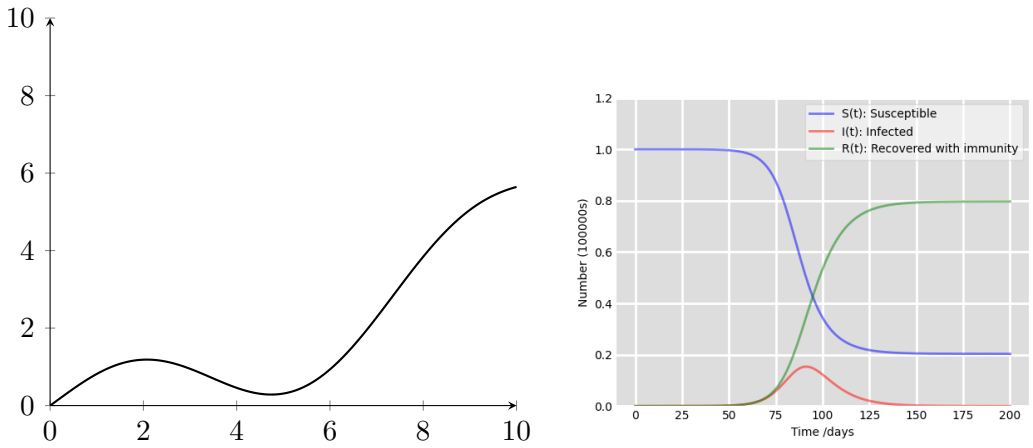


Figure 1: Left: A graph can be used to study changing quantities. Right:the SIR model.

- Mechanics, the equation of motion $F(x,t) = m\frac{d^2x}{dt^2} (= ma)$
- Electrodynamics, thermodynamics, fluid mechanics (Mathematical Analysis II)
- Statistics, machine learning (Mathematicala Analysis II, Linear Algebra)
- Epidemiology (the SIR model $\frac{dS}{dt} = -\frac{\beta S(t)I(t)}{N}$, $\frac{dI}{dt} = \frac{\beta S(t)I(t)}{N} - \gamma I(t)$, $\frac{dR}{dt} = \gamma I(t)$)

Mathematical Analysis I

Summary of the course

- properties of real numbers, concept of sets.
- mathematical induction. the summation notation.
- functions. limit of functions, continuity.
- trigonometric functions ($\cos x, \sin x$), exponential function e^x, logarithmic function $\log x$.
- differential calculus and applications.
- Taylor's formula, approximation of functions
- integral of functions. the relation between integration and differentiation.
- basic differential equations.
- numerical seqeunces and series, complex numbers

Reviews of high school mathematics

We will define the following material in a more formal way, but it is important to be familiar with practical calculations and methods of solving basic questions.

See self-assessment test

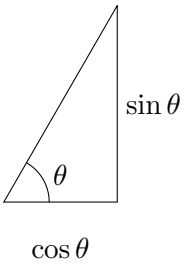


Figure 2: The trigonometric functions of θ.

Equations, inequalities

For example, the formula $(x-1)^2 = x^2 - 2x + 1$ is obtained by expanding the left-hand side. Such a formula is called an **identity**. $x^2 + 3x + 2 = 0$ does not hold for all x, but only for $x = -2, -1$. Such a formula is called an **equation**, and $-2, -1$ are called **solutions** to that equation. $5 - \frac{x}{2} > 2$ is satisfied for $x < 6$. This is called an **inequality** and all numbers $x < 6$ are called **solutions**.

Roots

$\sqrt{2}$ is the number whose square is 2. Its value is $\sqrt{2} = 1.41421356....$

For any positive number x, there is the square root $\sqrt{x}$, such that $(\sqrt{x})^2 = x$.

Similarly, for any natural number n, the n-th root $\sqrt[n]{x}$, such that $(\sqrt[n]{x})^n = x$. $\sqrt{16} = 4$, $\sqrt[3]{8} = 2$.

Solve $\sqrt[3]{x-1} = 3$, $\sqrt{x^2-5} > 2$.

Trigonometry

Consider the right triangle having one angle θ. The trigonometric functions $\sin\theta, \cos\theta$ are defined as in the figure.

It holds that $\cos 45^\circ = \frac{1}{\sqrt{2}}, \sin 45^\circ, \frac{1}{\sqrt{2}}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \sin 30^\circ = \frac{1}{2}, \cos 60^\circ = \frac{1}{2}, \sin 60^\circ = \frac{1}{\sqrt{2}}, \cos 90^\circ = 0, \sin 90^\circ = 1$.

Exercise: solve $\sin^2(3x) + \sin(3x) - 2 = 0$. Hint: put $\sin(3x) = X$.

Exp, log

Let $a > 0$. For any (real) number x, there is a^x. $2^3 = 8, 5^{\frac{1}{2}} = \sqrt{5}, 3^{\frac{1}{3}} = \sqrt[3]{3}$.

Let $a > 0$, $x > 0$. $\log_a x$ is the number such that $a^{\log_a x} = x$.

$\log_2 8 = 3, \log_3(\sqrt{3}) = \frac{1}{2}, \log_2(\frac{1}{2}) = -1$.

Graphs

Let $f(x)$ be a function, that is an assignment of a number $f(x)$ to x. Its graph is obtained by collecting the points $(x, f(x))$.

Show the point $(1, 2), (2, -3)$ on the plane.

Sketch the graphs $f(x) = \frac{1}{2}x + 3, x^2 + 3x + 4, \sin(x), 2^x, \frac{1}{x}, \log_3 x$.

Sep 25. Mathematical logic, symbols, sets

Mathematical logic

Mathematical reasoning proceeds as a sequence of mathematical statements. Mathematical statements are expressions about mathemathcal objects that are either true or false. $2 + 2 = 4$

is a true statement, while $2+2=5$ is a false statement. Random combinations of words and symbols are not necessarily a statement, for example $2=(8+1(3-$.

From one (or more) statement(s), we can form compound statements. For example,

- "$2+2=4$ and $2+2=5$" is a stametent, but it is false.
- "$2+2=4$ or $2+2=5$" is a true statement.
- not "$2+2=5$", often written as $2+2\neq 5$, is a true statement.
- "If $2+2=4$, then $2+2=5$" is a false statement.
- "If $2+2=5$, then $2+2=4$" is a true statement (because the hypothesis is false).

One can also consider for many things together for example,

- "For all real number x, $x^2\geq 0$" is a true statement.
- "There exist a real number x such that $x^2<0$" is a false statement.

We can also compound already compound statements. For example,

- not "For all real number x, $x^2\geq 0$" means that "there exist a real number x such that not $x^2\geq 0$". This is false.
- not "There exist a real number x such that $x^2<0$" means that "for all real number x, not $x^2<0$". This is true.

Mathematical symbols

We use **symbols** for general mathematical objects. Before using a symbol, we **declare** what kind of object it is. For example,

- x,y,z,a,b,c often for numbers (but be careful of the declaration)
- f,g,h,F,G,H often for "functions" (which we will study later)

A symbol might be "recycled", that is, can be declared to be something different (unfortunately, we have only 26×2 alphabets).

Symbols are very useful because we can express general properties of certain mathematical objects at the same time, without specifying them every time.

Integers and rational numbers

We assume that we know

- **integers**: $0,1,2,3,\cdots,100,101,\cdots,492837498\cdots,-1,-2,-3,\cdots$
- **rational numbers**: $\frac{1}{2},\frac{2}{3},\cdots,\frac{23}{62518},-\frac{3028746}{26543},\cdots$ (integers are also rational numbers)
- calculations between them (sum, difference, product, division, order)

On rational numbers, we have the set of operations $+$ (summation), $\cdot$ (product): For x,y,z rational numbers (**declaration**), $x+y$ and $x\cdot y$ are again rational numbers and they satisfy

- (commutativity) $x+y=y+x, x\cdot y=y\cdot x$
- (associativity) $(x+y)+z=x+(y+z), (x\cdot y)\cdot z=x\cdot(y\cdot z)$
- (distributive law) $(x+y)\cdot z=xz+yz$

- (zero and unity) There are special distinct rational numbers, called 0 and 1, such that $x + 0 = x$ and $x \cdot 0 = 0$. And $x \cdot 1 = x$.
- (negative) There is only one rational number, which we call $-x$, such that $x + (-x) = 0$.
- (inverse) If $x \neq 0$, there is only one rational number, which we call x^{-1}, such that $x \cdot x^{-1} = 1$.

We often simply write xy for $x \cdot y$ and $x - y$ for $x + (-y)$. xy^{-1} is also written as $\frac{x}{y}$.

Exercises Take concrete rational numbers and check these properties!

Other properties of rational numbers can be **derived** from these. Indeed, we can prove the following[1]

Theorem 1. *Let a, b, c, d be rational numbers.*

- *if $a + b = a + c$, then $b = c$.*
- *$-(-a) = a$.*
- *$a(b - c) = ab - ac$.*
- *$a \cdot 0 = 0 \cdot a = 0$.*
- *if $ab = ac$ and $a \neq 0$, then $b = c$.*
- *if $a \neq 0$, then $a^{-1} \neq 0$ and $(a^{-1})^{-1} = a$.*
- *if $ab = 0$, then $a = 0$ or $b = 0$.*
- *$(-a)b = -(ab)$ and $(-a)(-b) = ab$.*
- *if $b \neq 0, d \neq 0$, then $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.*
- *if $b \neq 0, d \neq 0$, then $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$.*
- *if $a \neq 0, b \neq 0$, then $(\frac{a}{b})^{-1} = \frac{b}{a}$.*

Proof. We only prove a few of them and **leave the rest as exercises**.

Let us assume that $a + b = a + c$. Then, we take $-a$ and

$$\begin{aligned}(a + b) + (-a) &= a + (b + (-a)) \text{ (associativity)} \\ &= a + ((-a) + b) \text{ (commutativity)} \\ &= (a + (-a)) + b \text{ (associativity)} \\ &= 0 + b \text{ (definition of 0)} \\ &= b \text{ (property of 0)}\end{aligned}$$

Similarly, $(a+c)+(-a) = c$. But as $a+b = a+c$, we have $b = (a+b)+(-a) = (a+c)+(-a) = c$.

Assume that $a \neq 0$. We show that $a^{-1} \neq 0$ by contradiction. Indeed, if we had $a^{-1} = 0$, then we would have $0 \cdot a = a^{-1} \cdot a = 1$, contradiction. Therefore, $a^{-1} \neq 0$ and $1 = a \cdot a^{-1} = a^{-1} \cdot a$, hence $a = (a^{-1})^{-1}$.

If $ab = 0$ and $a \neq 0$, then we can take a^{-1} and $0 = a^{-1}0 = a^{-1}ab = 1 \cdot b = b$.

□

[1] statements that can be proven are called **theorems**, and the properties that we assume are called **axioms**.

Sep 26. Concept of sets, set of numbers

Order in rational numbers

There is also an **order relation** $<$ ("x is larger than y": $y < x$) which satisfies, for x, y, z rational,

- if $0 < x, 0 < y$, then $0 < xy$ and $0 < x + y$.
- if $x < y$, then $x + z < y + z$.
- if $x \neq 0$, either $0 < x$ or $x < 0$ but not both.
- It is not true that $0 < 0$.

$x < y$ and $y > x$ have the same meaning.

We say that x is **positive** if $0 < x$ and x is **negative** if $x < 0$. If x is not positive, then either $x = 0$ or $x < 0$ and in this case we say x is **nonpositive** and write $x \leq 0$ (again, $x \geq 0$ and $0 \leq x$ mean the same thing). Similarly, if $x > 0$ or $x = 0$, we say x is **nonnegative** and write $x \geq 0$.

In addition to the "axioms", we also use the logic that, if an equality or inequality holds for some x and if $x = y$, then it also holds for y. **Example**: if $x < z$ and $x = y$, then $y < z$.

Notation: for any number x, we write $x^2 = x \cdot x$. Similarly, $x^3 = x \cdot x \cdot x$, and so on.

With the properties above, we can prove the following.

Theorem 2. *Let a, b, c, d rational numbers. Then*

- *$a < b$ if and only if $a - b < 0$.*
- *one and only one of the following holds: $a < b, b < a, a = b$.*
- *if $a < b, b < c$ then $a < c$.*
- *if $a < b$ and $c > 0$, then $ac < bc$.*
- *if $a \neq 0$, then $a^2 (= a \cdot a) > 0$.*
- *$1 > 0$.*
- *if $a < b$ and $c < 0$, then $ac > bc$.*
- *if $a < b$, then $-a > -b$.*
- *if $ab > 0$, then either $a > 0, b > 0$ or $a < 0, b < 0$.*
- *if $a < c$ and $b < d$, then $a + b < c + d$.*

Proof. We only prove a few of them and leave the rest as exercises. In general, to show "A if and only if B", it is enough to show that "if A, then B" and "if B, then A", this is because "A only if B" implies that "if not B, then not A", and by contradiction, "if A, not not B", but "not not B" means B.

If $a < b$, then by adding $-b$ to both sides, we get $a - b < 0$. Conversely, if $a - b < 0$, by adding b to both sides we get $a < b$.

If $a = b$, then $b - a = 0$ and we know that both $b - a > 0$ and $b - a < 0$ are false and hence $b > a$ and $b < a$ are false. If $a < b$, then $a - b < 0$ and $a - b = 0$ is false, and hence $a = b$ is false.

If $a < b$, then $0 < b - a$ and $0 < c \cdot (b - a) = bc - ac$, hence $ac < bc$.

If $a \neq 0$, then either $a > 0$ or $a < 0$. For the case $a > 0$, we have $a^2 = a \cdot a > 0$. For the case $a < 0$, we have $-a > 0$ and $a^2 = (-a)^2 > 0$.

□

All these "theorems" about rational numbers should be well-known to you. But it is important that we could prove them from a few axioms, which we assume to be true.

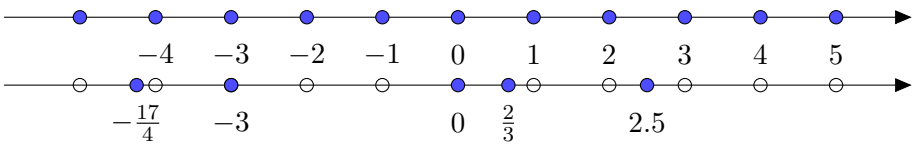


Figure 3: A geometric representation of integers and rational numbers.

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Exercises Check the remaining statements.

Integers and rational numbers can be represented on a line.

Naive set theory

It is very often convenient to consider sets of numbers. For example, we may consider the set $\mathbb{Q}_+$ of positive rational numbers, or the set of multiples of 2, and so on. In mathematics, a **set** is a collection of mathematical objects. The most precise treatment of sets requires a theory called axiomatic set theory, but in this lecture we think of a set simply as a collection of known objects.

We often use capital letters $A, B, C, \cdots$ for sets, but in any case we declare that a symbol is a set. For a set S, we denote by $x \in S$ the statement "x is an **element** of S". We have already seen examples of sets: let us give them special symbols.

- $\mathbb{Q}$: the set of rational numbers
- $\mathbb{Z}$: the set of integers

In general, we can consider two ways of constructing sets.

- By nomination. We can nominate all elements of a set. For example, $A = \{0, 1, 2, 3\}$ and $B = \{1, 10, 100, 1000\}$ are sets.
- By specification. We include all elements of an existing set with specific properties. For example, $A = \{x \in \mathbb{Z} : \text{ there is } y \in \mathbb{Z} \text{ such that } x = 2y\}$ (read that "A is the set of integers such that there is an integer y such that $x = 2y$") is the set of multiples of 2 (we recycled the symbol A. When we do this, we shall always declare it).

For a set constructed by nomination, the order and repetition do not matter: $\{0, 1, 2, 3\} = \{0, 3, 2, 1\} = \{0, 0, 1, 1, 1, 2, 3\}$. In other words, a set is defined by its elements.

A construction by specification appears very often. Let us introduce a more symbol.

- $\mathbb{N} = \{x \in \mathbb{Z} : x > 0\}$ is called the set of **natural numbers**.
- $\emptyset$ is the set that contains nothing and called the **empty set**. $\emptyset$ is a subset of any set: if A is a set, the statement "if $x \in \emptyset$ then $x \in A$" is satisified just because there is no such x!

Subsets

Let B be a set. We say that A is a **subset** of B if all elements of A belong to B, and denote this by $A \subset B$. It holds that $A \subset A$ for any set A.

Example 3.

- Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 3, 4\}$. Then $A \subset B$.
- $\mathbb{N} \subset \mathbb{Z}$.
- Let $A = \{1, 2, 3, 4, 5, 6\}$. Then $A \subset \mathbb{N}$.

It may happen that $A \subset B$ and $B \subset A$, that is, all elements of A belong to B and vice versa. This means that A and B are the same as sets, and in this case we write $A = B$.

The definition by specification $A = \{x \in B : x \text{ satisfies the property XXX...}\}$ gives always a subset, in this case of B. Note also that x in this definition has no meaning ("dummy"). One can write it equivalently $A = \{y \in B : y \text{ satisfies the property XXX...}\}$.

For $x \in A$, the set $\{x\}$ that contains only x should be distinguished from x. It is a subset of A: $\{x\} \subset A$.

Unions, intersections, complements

If A and B are sets, then we can consider the set which contains the elements of A and B, and nothing else. It is called the **union** of A and B and denoted by $A \cup B$.

Example 4. • Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 3, 4\}$. Then $A \cup B = \{0, 1, 2, 3, 4\}$.

Similarly, we can consider the set of all the elements which belong both to A and B, and nothing else. It is called the **intersection** of A and B and denoted by $A \cap B$.

Example 5. • Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 3, 4\}$. Then $A \cap B = \{1, 3\}$.

Furthermore, the **difference** of B with respect to A is all the element of A that do not belong to B and is denoted by $A \setminus B$ (note that this is different from $B \setminus A$).

Example 6. • Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 3, 4\}$. Then $A \setminus B = \{2\}$.

We can consider the union of more than two sets: $A \cup (B \cup C)$. By considering the meaning, this set contains all the elements which belong either A or $B \cup C$, which is to say all elements which belong either A or B or C. Therefore, the order does not matter and we can write $A \cup B \cup C$. Similarly, $A \cap B \cap C$ is the intersection of A, B and C.

We may consider a **family of sets** $\{A_i\}_{i \in I}$ **indexed by another set** I. For example, we can take $\mathbb{N}$ as the index set and $A_n = \{m \in \mathbb{Z} : m \text{ is a multiple of } n\}$. For a family of set, we can define the union and the intersection analogously and we denote them by

$$\bigcup_{i \in I} A_i, \qquad \bigcap_{i \in I} A_i,$$

respectively.

Sep 27. Sets and logic

Sets by specification

Let us recall that, if we fix a set A, we can define a subset of A by specification: it is the subset of elements x of A that satisfy a certain condition $\varphi(x)$:

$$\{x \in A : \varphi(x)\},$$

where $\varphi(x)$ is a condition on x. For example, $\{x \in \mathbb{Z} : x > 10\}$ is the set of integers larger than 10.

On one hand, we can consider the combined conditions: for example, the condition that $x > 10$ and the condition that $x < 15$ can be considered at the same time. The set of integers that satisfy both of the condition is

$$\{x \in \mathbb{Z} : x > 10 \text{ and } x < 15\} = \{11, 12, 13, 14\}.$$

On the other hand, we observe that this set is the intersection of two sets:

$$\{x \in \mathbb{Z} : x > 10\} = \{10, 11, 12, 13, 14, 15, 16, 17, \cdots\},$$
$$\{x \in \mathbb{Z} : x < 15\} = \{-2, -1, 0, 1, \cdots, 10, 11, 12, 13, 14, 15\}.$$

This can be generalized as follows. If $B = \{x \in A : \varphi(x)\}, C = \{x \in A : \psi(x)\}$, then $B \cap C = \{x \in A : \varphi(x) \text{ and } \psi(x)\}$.

Similarly, the union of two sets is related with "or" as follows. For example,

$$B = \{x \in \mathbb{Z} : x > 15\} = \{16, 17, 18, \cdots\},$$
$$C = \{x \in \mathbb{Z} : x < 10\} = \{\cdots - 2, -1, 0, 1, \cdots, 8, 9\}.$$

We observe that $B \cup C = \{\cdots - 2, -1, 0, 1, \cdots, 8, 9, 16, 17, 18, \cdots\}$, which is $B \cup C = \{x \in \mathbb{Z} : x > 15 \text{ or } x < 10\}$. In general, if $B = \{x \in A : \varphi(x)\}, C = \{x \in A : \psi(x)\}$, then $B \cup C = \{x \in A : \varphi(x) \text{ or } \psi(x)\}$.

Let us the consider the negation. For example,

$$B = \{x \in \mathbb{Z} : x > 15\} = \{16, 17, 18, \cdots\}.$$

Because the negation of $x > 15$ is $x \leq 15$, we have

$$\mathbb{Z} \setminus B = \{\cdots - 2, -1, 0, 1, \cdots, 14, 15\} = \{x \in \mathbb{Z} : x \leq 15\}.$$

In general, if $\neg\varphi(x)$ is the negation of $\varphi(x)$, then it holds that, for $B = \{x \in A : \varphi(x)\}$, $A \setminus B = \{x \in A : \neg\varphi(x)\}$.

Let us consider the set $\{x \in \mathbb{Q} : (x-1)(x-3) > 0\}$. To understand better this set, we need to understand the condition $(x-1)(x-3) > 0$. The left-hand side is a product of two rational numbers. The product of two rational numbers is positive if and only if one of the following cases is true.

- $x - 1 > 0$ and $x - 3 > 0$
- $x - 1 < 0$ and $x - 3 < 0$.

They are further equivalent to

- $x > 1$ and $x > 3$
- $x < 1$ and $x < 3$.

Note that $x > 1$ is true if $x > 3$. Similarly, $x < 3$ is true if $x < 1$. Therefore, these conditions are equivalent to

- $x > 3$
- $x < 1$

Altogether, we have

$$\{x \in \mathbb{Q} : (x-1)(x-3) > 0\} = \{x \in \mathbb{Q} : x > 3 \text{ or } x < 1\} = \{x \in \mathbb{Q} : x > 3\} \cup \{x \in \mathbb{Q} : x < 1\}.$$

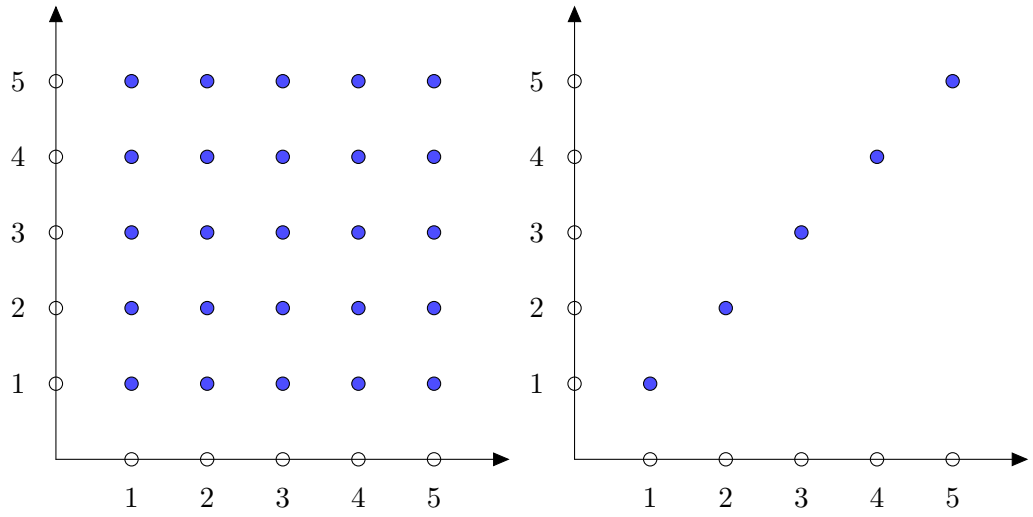


Figure 4: Left: the set of all ordered pairs $\{1,2,3,4,5\} \times \{1,2,3,4,5\}$. Right: a subset $\{(1,1),(2,2),(3,3),(4,4),(5,5)\} \subset \{1,2,3,4,5\} \times \{1,2,3,4,5\}$.

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The set of subsets, the set of pairs, graphs

We can consider also certain **sets of sets**.

Example 7.

- $\{1,2,3\},\{2\},\{1,4,6,7\}$ are sets. We can collect them together

$$\{\{1,2,3\},\{2\},\{1,4,6,7\}\}.$$

  This is a set of sets. It is different from the set of their elements $\{1,2,3,4,6,7\}$.

- Let $A = \{1,2,3\}$. We can collect all subsets of A:

$$\{\emptyset,\{1\},\{2\},\{3\},\{1,2\},\{1,3\},\{2,3\},\{1,2,3\}\}.$$

- One can also consider the set of all subsets of $\mathbb{N},\mathbb{Z},\mathbb{Q}$, but we cannot name all the elements: they are infinite. For example, for $\mathbb{N} = \{1,2,3,4,\cdots\}$, the set of subsets of $\mathbb{N}$ is infinite.

For sets A, B, we can consider **ordered pairs** of elements in A and B.

Example 8.

- Let $A = \{1,2,3\}, B = \{3,4\}$. Then the sef $A \times B$ of the ordered pairs of A, B is

$$A \times B := \{(1,3),(2,3),(3,3),(1,4),(2,4),(3,4)\}.$$

- If we take $\mathbb{N}$, then $\mathbb{N} \times \mathbb{N}$ is the set of all ordered pairs of natural numbers. $\mathbb{N} \times \mathbb{N} = \{(1,1),(1,2),(2,1),(1,3),(2,2),(3,1),\cdots\}$.

Ordered pairs can be described using **graphs**. If $A, B \subset \mathbb{Z}$ have finitely many points, say m, n respectively, then there are $m \cdot n$ ordered pairs. We take the horizontal axis for A and the vertical axis for B.

To obtain the graph of $A \times B$, we should mark the point (x,y) if and only if $x \in A$ and $y \in B$. For any subset X of $A \times B$, we should mark the point (x,y) if and only if $(x,y) \in X$. See Figure 4.

The graph of a relation can be understood in terms of ordered pairs. Let $A = \{1,2,3,4,5,6\}$, and $B = \{(x,y) \in A \times A : y = 2x\}$. Let us give all elements of B and draw its graph. We check all $6 \times 6 = 36$ elements. See Figure 5, it is $\{(1,2),(2,4),(3,6)\}$. Notice that it is on a **straight line**!

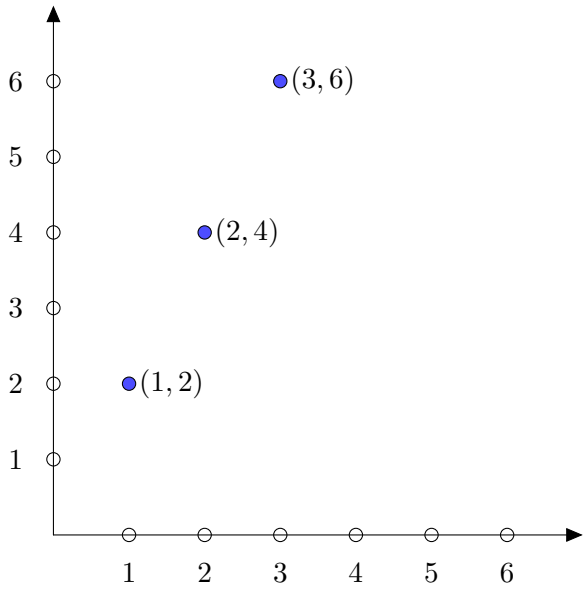


Figure 5: The set of all ordered pairs $(x, y) \in \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$ with $y = 2x$.

Oct 2. Real numbers

Are rational numbers all we need?

It is true that, in the real world, we can measure quantities to a certain accuracy, so we get numbers in a decimal representation:

- $c = 299792458[\mathrm{m} \cdot \mathrm{s}^{-1}]$ (the speed of light)
- $G = 0.0000000000667430(15)[\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}]$ (the gravitational constant), where (15) means these digits might be incorrect.
- Any other measured quantity in the real world.

And any experiment has a certain accuracy, so it makes sense only to write a certain number of digits, so rational numbers seem to suffice.

But for certain cases, we know that we should consider **irrational numbers**. For example,

- $\sqrt{2} = 1.41421356\cdots$, the number x such that $x^2 = 2$.
- $\pi = 3.1415926535\cdots$, the circumference of the circle with diameter 1.
- $e = 2.718281828\cdots$, Napier's number (we will define it in the lecture).
- Any decimal number which is not repeating.

For the next theorem, we need a **proof by contradiction**: by assuming the converse of the conclusion, we derive a contradiction, then we can conclude that the converse of the conclusion is false, that is, the conclusion is correct.

Recall that an integer p is **even** if it is a multiple of 2 (there is another integer r such that $p = 2r$), and p is **odd** if it is not even.

Theorem 9. *$\sqrt{2}$ is not a rational number.*

Proof. We prove this by contradiction, that is, we assume that $\sqrt{2}$ is a rational number. So there are integers p, q such that $\sqrt{2} = \frac{p}{q}$. We may assume that this is already reduced (that is, not a fraction like $\frac{4}{8}$ but like $\frac{1}{2}$. It's the form which you cannot simplify further).

As $\sqrt{2} \cdot \sqrt{2} = 2$, we have $\frac{p}{q} \cdot \frac{p}{q} = \frac{p^2}{q^2} = 2$, hence $p^2 = 2q^2$. As $\frac{p}{q}$ is reduced, there are two cases.

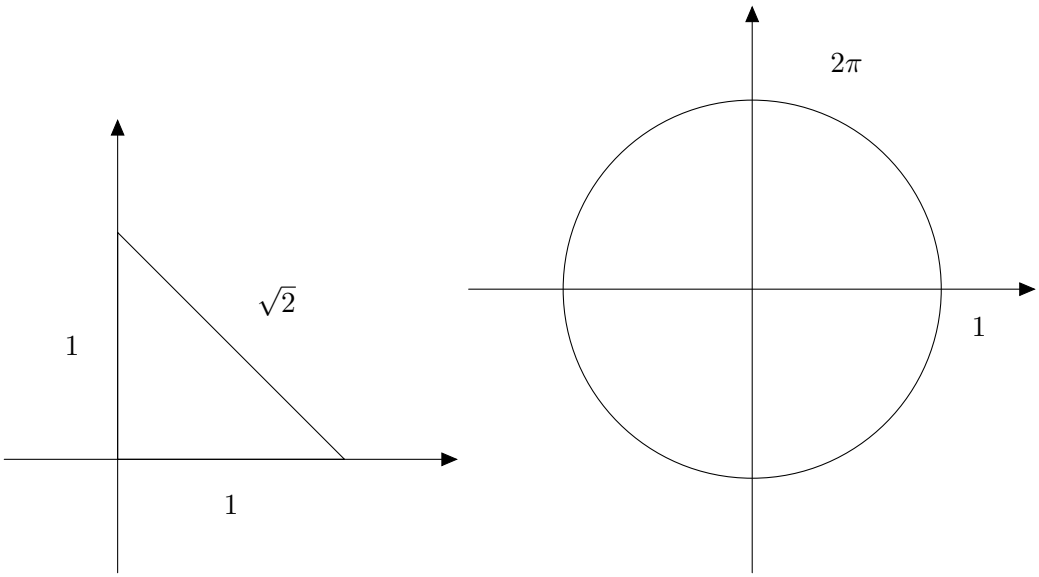


Figure 6: Left: the right triangle with equal sides 1. By the Pytagoras' theorem, the longest side is $\sqrt{1^2+1^2} = \sqrt{2}$. Right: the unit circle with radius 1 (diameter 2). The length of the circle (the circumference) is 2π.

- if p is odd, then the equality $p^2 = 2q^2$ is even = odd, contradiction.
- if p is even, then q is odd and we can write is as $p = 2r$, with another integer r, and $p^2 = 4r^2 = 2q^2$, and $2r^2 = q^2$. This is even = odd, contradiction.

So, in all cases we arrived a contradiction from the assumption that $\sqrt{2}$ is rational. This means that $\sqrt{2}$ is irrational. □

Exercise. Prove that $2\sqrt{2}$ is irrational.

It has been proven that π and e are irrational, but they are more difficult. Instead, it can be easily proven that any nonrepeating decimal number cannot be rational. This means *there are many irrational numbers.*

In other words, the set of rational numbers have "many spaces between them". We should fill them in with irrational numbers, so that the set of real numbers is a "continuum".

The axioms of the real numbers

Here we start the study of Mathematical Analysis, based on the set of **real numbers**. Our approach is synthetic, in the sense that we take the axioms for real numbers for granted, and develop the theory on them. It is also possible to "costruct" real numbers from rational numbers, and rational numbers from integers, integers from natural numbers, and so on, but at some point we have to assume certain axioms for simpler objects. If you are interested, look at "Dedekind's cut" (for real numbers), or "Peano's axioms" (for natural numbers).

We assume that, the set $\mathbb{R}$ of real numbers is equipped with operations $+$ (summation), $\cdot$ (product) and for x, y, z real numbers, $x + y$ and $x \cdot y$ are again real numbers and they satisfy (*just the same properties for rational numbers* $\mathbb{Q}$)

- (commutativity) $x + y = y + x, x \cdot y = y \cdot x$
- (associativity) $(x + y) + z = x + (y + z), (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- (distributive law) $(x + y) \cdot z = xz + yz$

- (zero and unity) There are special distinct rational numbers, called 0 and 1, such that $x + 0 = x$ and $x \cdot 0 = 0$. And $x \cdot 1 = x$.
- (negative) There is only one real number, which we call $-x$, such that $x + (-x) = 0$.
- (inverse) If $x \neq 0$, there is only one real number, which we call x^{-1}, such that $x \cdot x^{-1} = 1$.

There is also an **order relation** $<$ which satisfies, for x, y, z real,

- if $0 < x, 0 < y$, then $0 < xy$ and $0 < x + y$.
- if $x < y$, then $x + z < y + z$.
- if $x \neq 0$, either $0 < x$ or $x < 0$ but not both.
- $0 \not< 0$

We can prove Theorems for real numbers corresponding to Theorems 1, 2. Therefore, the real numbers have the same properties as the rational numbers, concerning the sum, product and order.

$\mathbb{R}$ includes $\mathbb{Z}$ and $\mathbb{Q}$: $1 \in \mathbb{R}$, hence $2 = 1 + 1, 3 = 1 + 1 + 1, \cdots$ and $-1, -2, \cdots \in \mathbb{R}$. Also, if $p, q \in \mathbb{Z}$, $\frac{p}{q} \in \mathbb{R}$.

We say that $S \subset \mathbb{R}$ is **bounded above** if there is $x \in \mathbb{R}$ such that for any $y \in S$ it holds that $y \leq x$, and we write $S \leq x$. S is said to be **bounded below** if there is $x \in \mathbb{R}$ such that for any $y \in S$ it holds that $y \geq x$, and we write $S \geq x$.

If S is both bounded above and below, we say that S is **bounded**.

If S is bounded above, then any $x \in \mathbb{R}$ such that $S \leq x$ is called an **upper bound** of S. Similarly, if $x \leq S$, then x is said to be a **lower bound** of S.

If S has a least upper bound, that is there is x such that $S \leq x$ and $x \leq y$ for any upper bound y of S, then x is called the **supremum** of S and we denote it by $x = \sup S$. Similarly, if S has a largest lower bound x, then it is called the **infimum** of S and we denote it by $x = \inf S$.

Intervals

In the set of real numbers, we can consider **intervals**: let $a, b \in \mathbb{R}$ and $a < b$. We introduce

- $(a, b) = \{x \in \mathbb{R} : a < x, x < b\}$ (an **open** interval)
- $(a, b] = \{x \in \mathbb{R} : a < x, x \leq b\}$
- $[a, b) = \{x \in \mathbb{R} : a \leq x, x < b\}$
- $[a, b] = \{x \in \mathbb{R} : a \leq x, x \leq b\}$ (a **closed** interval)
- $(a, \infty) = \{x \in \mathbb{R} : a < x\}$
- $[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$
- $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$
- $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$

Remember that, a, b are given numbers, and x is a "dummy" number. You can write them in a different way, without using x:

- (a, b) is the set of all numbers larger than a and smaller than b
- $[a, b]$ is the set of all numbers larger than or equal to a and smaller than or equal to b

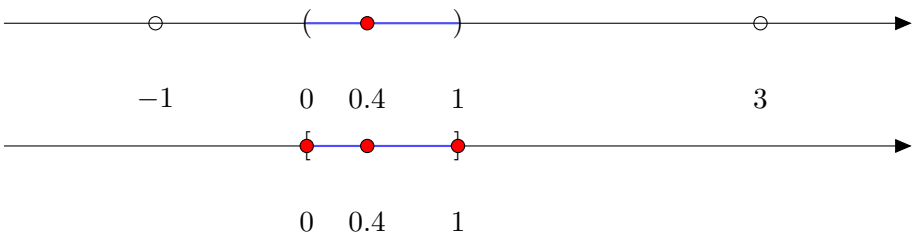


Figure 7: Open and closed intervals $(0,1)$ and $[0,1]$. The open interval does not include the edges $0, 1$, while the closed interval $[0,1]$ does.



Figure 8: The set S approximating $\sqrt{2}$, which is bounded by 1.5.

Example 10. Consider $(0,1)$.

- $0.1, 0.2, 0.5, 0.999 \in (0,1)$.
- $0, 1, 2, 3, 10, -1, -2 \notin (0,1)$.
- $\sup(0,1) = 1$.
- $\inf(0,1) = 0$.

Consider $[0,1]$.

- $0, 0.1, 0.2, 0.5, 0.999, 1 \in [0,1]$.
- $2, 3, 10, -1, -2 \notin [0,1]$.
- $\sup[0,1] = 1$.
- $\inf[0,1] = 0$.

The axion of sup

What distinguishes $\mathbb{R}$ from $\mathbb{Q}$ is the following.

- (the least upper bound axiom, or the completeness axiom) every nonempty subset S of $\mathbb{R}$ which is bounded above has a supremum: there is $B \in \mathbb{R}$ such that $B = \sup S$.

This should imply that $\sqrt{2} = 1.41421356\cdots$ belongs to $\mathbb{R}$! Indeed, let us take, by chopping the digits of $\sqrt{2}$, $S = \{1, 1.4, 1.41, 1.414, 1.4142, \cdots\}$. S is bounded above, indeed, $1.5 > 1, 1.4, 1.41, 1.414, \cdots$. On the other hand, if x has a decimal representation, e.g. 1.415, then there is a smaller number $x' = 1.4149$. So, $\sup S$ should be exactly $\sqrt{2}$. We will see this more precisely later.

(A lemma is a theorem (a consequence of axioms) used to prove a more important theorem)

Lemma 11. *Suppoese that $S \subset \mathbb{R}$ is bounded above and nonempty and $B = \sup S$. Then, for any $\epsilon > 0$, there is $x \in S$ such that $B - \epsilon < x$.*

Proof. By contradiction, assume that there is $\epsilon > 0$ such that $B - \epsilon \geq x$ for all $x \in S$. Then $B - \epsilon$ is an upper bound of S. Clearly it holds that $B - \epsilon < B$. This contradicts the definition of $\sup S = B$, the smallest upper bound of S. □

Theorem 12 (Archimedean property)**.** *The set* $\mathbb{N} = \{1, 2, 3, \cdots\}$ *is not bounded above.*

Proof. By contradiction, assume that $\mathbb{N}$ were bounded above. Then by the completeness axiom, there is $B = \sup \mathbb{N}$. By the lemma above, for $\epsilon = \frac{1}{2}$, there is $n \in \mathbb{N}$ such that $B - \frac{1}{2} < n$. But then $B < n + \frac{1}{2} < n + 1 \in \mathbb{N}$, and this contradicts the assumption that B were the upper bound of $\mathbb{N}$. This implies that $\mathbb{N}$ is not bounded above. □

(A corollary is a theorem which follows easily from a more complicated theorem)

Corollary 13. *For any* $x \in \mathbb{R}$*, there is* $n \in \mathbb{N}$ *such that* $x < n$*. For any* $y, z \in \mathbb{R}$ *and* $z > y$*, there is* $n \in \mathbb{N}$ *such that* $\frac{1}{n} < z - y$*.*

Proof. By the theorem above, x is not an upper bound of $\mathbb{N}$, so there is n such that $x < n$. By applying this to $\frac{1}{z-y}$, there is n such that $\frac{1}{z-y} < n$, which implies that $\frac{1}{n} < z - y$. □

Therefore, we can represent the set of real numbers by a straight line, and any point $x \in \mathbb{R}$ is on the line and it falls between an integer n and another $n - 1$ (possibly $x = n$). Conversely, any point on the line gives an element in $\mathbb{R}$.

Oct 3. Irrationality of the square root of 2, Peano's axioms.

Note that $\mathbb{Q}$ does not have the completeness property!

Theorem 14. *Let* $A = \{x \in \mathbb{R} : x^2 < 2\} \subset \mathbb{R}$*. Then* A *is bounded above, and* $s^2 = 2$*, where* $s = \sup A$*.*

Proof. A is bounded above, indeed, if $x^2 < 2$, then $x^2 < 4 = 2^2$, and hence $x < 2$ (try proving this by contradiction and using the fact that $0 < a < b$ implies $a^2 < b^2$).

$1 \in A$, so A is nonempty. Let $s = \sup A \in \mathbb{R}$. Then $s > 0$ because $1 \in A$, thus $s \geq 1 > 0$.

We prove $s^2 = 2$ by contradiction.

- if $s^2 < 2$, then we take $\epsilon > 0$ such that $0 < \epsilon < \frac{2-s^2}{s}$ (or $s\epsilon < 2 - s^2$) and $\epsilon < s$ (this is possible by Corollary 13). Then $(s + \frac{\epsilon}{4})^2 = s^2 + s\frac{\epsilon}{2} + \frac{\epsilon^2}{16} < s^2 + \frac{s\epsilon}{2} + \frac{s\epsilon}{2} < s^2 + s\epsilon < 2$, so $s + \frac{\epsilon}{4} \in A$. Therefore, s is not an upper bound of A (because $s + \frac{\epsilon}{4} \in A$), contradiction.
- if $s^2 > 2$, then we take $\epsilon > 0$ such that $0 < \epsilon < \frac{s^2-2}{s}$ (or $s\epsilon < s^2 - 2$) and $\epsilon < s$. Then $(s - \frac{\epsilon}{4})^2 = s^2 - s\frac{\epsilon}{2} + \frac{\epsilon^2}{16} > s^2 - s\epsilon > 2$. This implies that $s - \frac{\epsilon}{4}$ is an upper bound of A (because for any $x \in A$, $x^2 < 2 < (s - \frac{\epsilon}{4})^2$, so $x < s - \frac{\epsilon}{4}$), therefore, s is not the least upper bound of A (because $s - \frac{\epsilon}{4} \in A$ is another upper bound, smaller than s), contradiction.

But we know that there is no rational number $s \in \mathbb{Q}$ such that $s^2 = 2$. Hence $s = \sup A \notin \mathbb{Q}$. □

This also says that $s = \sqrt{2}$ belongs to $\mathbb{R}$.

For any $n \in \mathbb{N}$, we can define the n-th root of any positive number a and we denote it by $a^{\frac{1}{n}}$. The existence can be proved similarly.

The Peano axioms

The set $\mathbb{N}$ of natural numbers can be caracterized by the Peano axioms:

- $1 \in \mathbb{N}$
- For every $n \in \mathbb{N}$, $n + 1 \in \mathbb{N}$
- For every $n \in \mathbb{N}$, $n + 1 \neq 1$ (this means $0 \notin \mathbb{N}$)
- Let $S \subset \mathbb{N}$. If $1 \in S$ and $n + 1 \in S$ for any $n \in S$, then $S = \mathbb{N}$.

In other words, $\mathbb{N}$ consists of 1 and all other numbers obtained by adding 1 repeatedly to 1, and that is all. This is the precise definition of $\mathbb{N}$.

Mathematical induction

With this characterization, we obtain the mathematical induction. Let $\varphi(n)$ be a set of theorems depending on $n \in \mathbb{N}$. If $\varphi(1)$ is true, and if we can prove $\varphi(n+1)$ from $\varphi(n)$, then $\varphi(n)$ is true for all natural numbers. Indeed, let $S = \{n \in \mathbb{N} : \varphi(n) \text{ is true } \}$. S is a subset of $\mathbb{N}$, $1 \in S$ and if $n \in S$, then $n+1 \in S$. From the Peano axioms, we have $S = \mathbb{N}$. In other words, $\varphi(n)$ holds for all $n \in \mathbb{N}$.

Example 15. $n^2 \geq 2n - 1$ for all n.

Indeed, we apply mathematical induction to $\varphi(n) =$ "$n^2 \geq 2n - 1$". With $n = 1$, we have $1 \geq 2 \cdot 1 - 1 = 1$.

If we assume that this holds for n, then $(n+1)^2 = n^2 + 2n + 1 \geq 2n - 1 + 2n + 1 = 4n = 2n + 2n \geq 2n + 1 = 2(n+1) - 1$, therefore, we proved $\varphi(n+1)$ from $\varphi(n)$. We can now conclude that $\varphi(n)$ is true for all $n \in \mathbb{N}$.

Exercise: prove that $n > 0$ for all $n \in \mathbb{N}$.

The well-ordering principle

First we need the following.

Lemma 16. *For all $m, n \in \mathbb{N}$, if $n < m$, then $m - n \in \mathbb{N}$.*

Proof. This is proved by a double induction. Let $\varphi(m, n) =$ "if $m > n$, then $m - n \in \mathbb{N}$". Let us first set $n = 1$, $m = 1$. In this case, $m > n$ is not true, so we do not have to prove anything.

Assume that $\varphi(m, 1)$ is true, that is, if $m > 1$, then $m - 1 \in \mathbb{N}$. To prove $\varphi(m+1, 1)$, assume that $m + 1 > 1$, but $m + 1 - 1 = m \in \mathbb{N}$. By induction, $\varphi(m, 1)$ is true for all $m \in \mathbb{N}$.

Let us put $\psi(n) =$ "for all $m \in \mathbb{N}$, if $n < m$, then $m - n \in \mathbb{N}$. We proved $\psi(1)$.

Let us prove $\psi(n)$ by induction. Assume that $\psi(n)$ is true and we prove $\psi(n+1)$. Assume that $n+1 < m$ and we prove $m - (n+1) \in \mathbb{N}$. This does not happen if $m = 1, 2$, thus these cases are ok. For $m > 2$, we have $m - 1 > n > 0$, and $m - 1 \in \mathbb{N}$ by $\varphi(m, 1)$. Then by $\varphi(m-1, n)$, $m - (n+1) = m - 1 - n \in \mathbb{N}$. that is, we proved $\varphi(m, n+1)$. □

Theorem 17. *Let $A \subset \mathbb{R}$ be nonempty. If A is bounded below, then $-A$ is bounded above and $\inf(A) = -\sup(-A)$, where $-A = \{x \in \mathbb{R} : -x \in A\}$.*

Proof. Assume that $a \leq A$. Then if $-x \in A$, then $a \leq -x$, thus $x \leq -a$, and $-A$ is bounded above by $-a$. In particular, $\inf(A)$ is an upper bound of $-A$.

We show that $-\inf(A) = \sup(-A)$. For another upper bound b of $-A$, we have $x \leq b$ for all x such that $-x \in A$, $-b$ is a lower bound of A. Thus $-b \leq \inf A$, or $-\inf A \leq b$. This implies that $-\inf A = \sup(-A)$. □

We combine a proof by contradiction and mathematical induction.

Theorem 18. *For any nonempty subset $S \in \mathbb{N}$, there is the smallest element in S. That is, there is $n \in S$ such that $n \leq m$ for all $m \in S$.*

Proof. Let us call **A** the assumption that S is not empty.

Let us assume the contrary, that S does not admit the smallest element (call this assumption **B**). It means that, for any $n \in S$, there is $m \in S$ such that $m < n$.

Let $T = \{n \in \mathbb{N} : m > n, \text{ for all } m \in S\}$. We show that $T = \mathbb{N}$ by induction.

- First, $1 \in T$. To prove this, assume that $1 \notin T$ (call this $\mathbf{C}_1$). Then, there must be $m \in S$ such that $m \leq 1$. This means $1 \in S$. But 1 is always the smallest element of any subset of $\mathbb{N}$, contradicting **B**. Therefore, $\mathbf{C}_1$ is false and we obtain $1 \in T$.

- Next, let $n \in T$ and we prove that $n+1 \in T$. Assume that $n+1 \notin T$ (call this $\mathbf{C}_n$). Then, there is $m \in S$ such that $m \leq n+1$, but since $n \in T$, it must hold that $n < m$. This means that $m = n+1$ by the previous lemma, and any $\ell \leq n$ does not belong to S. Therefore, $m = n+1$ would be the smallest element of S, contradicting **B**. Therefore, $\mathbf{C}_n$ is false and we obtain $n+1 \in T$.

Then by induction (the Peano axioms) we have $T = \mathbb{N}$. This implies that for any $m \in S$ it it holds for $m < n$ for all $n \in T = \mathbb{N}$. But there is no such m (larger than any natural number), hence $S = \emptyset$. This contradicts the assumption **A** of the theorem. Therefore, the assumption **B** made in the proof is wrong. That is, S admits the smallest element. □

Alternatively, this can be proved as follows, but using the axiom of the least upper bound (the proof above uses only the Peano axioms). As $\mathbb{N}$ is bounded below, S is bounded below as well. Let $a = \inf S$. We show that $a \in S$. If not, for $\epsilon = \frac{1}{2}$, there is $m \in S$ such that $m < a + \frac{1}{2}$. If $m = a$, then $a \in S$ and it is the smallest element, If not, there is $n \in S$ such that $n < a + (m - a) = m$. By the previous lemma, $m - n \in \mathbb{N}$, but $a < n < m < a + \frac{1}{2}$, thus $m - n < \frac{1}{2}$, contradiction.

Corollary 19. *Let $x \in \mathbb{R}, x > 1$. There is $n \in \mathbb{N}$ such that $n - 1 \leq x < n$.*

Proof. By the Archimedean principle, there is n such that $x < n$. Therefore, the set $S = \{m \in \mathbb{N} : x < m\}$ is nonempty, and by the well-ordering principle, it has the smallest element n. As this is the smallest element, $n - 1 \leq x$ (otherwise $n - 1$ would be in S, because $1 < x < n$ thus $2 \leq n$ and $n - 1 \in \mathbb{N}$, contradicting the conclusion that n was the smallest of S). □

We have used this property before to find the decimal representation of x.

Oct 4. Natural numbers and induction.

Decimal representation of real numbers

We denote $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

We show that any real number $\mathbb{R}$ has a decimal representation: that is, any (positive) real number $x \in \mathbb{R}$ can be written in the form $x = a_0 + \frac{a_1}{10} + \frac{a_2}{100} + \cdots$, where a_0 is an integer and $a_1, a_2, \cdots$ are integers between 0 and 9 (negative numbers can be most commonly written as $-\sqrt{2} = -1.41421\cdots$, although an analogous representation can apply to negative numbers).

Examples:

- $\frac{1}{3} = 0.33333\cdots$
- $\sqrt{2} = 1.41421\cdots$
- $\pi = 3.14159\cdots$

Indeed, let $x \in \mathbb{R}$ be a real number and $x > 0$. By the Archimedean property, there is a natural number $n \in \mathbb{N}_0$ such that $n - 1 \leq x < n$ (this is possible, because any subset of $\mathbb{N}$ has the minimal element, which we prove below). We take $a_0 = n - 1$.

Note that $0 \leq x - a_0 < 1$. Therefore, $0 < 10(x - a_0) < 10$. Take $a_1 \in \mathbb{N}_0$ the largest natural number such that $a_1 \leq 10(x - a_0)$. As it is the largest, we have again $0 < 10(x - a_0) - a_1 < 1$.

We can repeat this procedure and obtain a_n, and it always hold that $x - a_0.a_1 \cdots a_n < 0.\underbrace{0\cdots01}_{n\text{-digits}}$.

Let $A = \{a_0, a_0.a_1, a_0.a_1a_2, a_0.a_1a_2a_3, \cdots\}$. This A is bounded (by $a_0 + 1$), hence it has the supremum s. Note that x is un upper bound of A, hence $\sup A \leq x$. On the other hand, if for any $\epsilon = 0.\underbrace{0\cdots01}_{n\text{-digits}}$, we have $x - \epsilon < a_0.a_1 \cdots a_n \in A$, therefore, $x \leq \sup A$. Altogether, $x = \sup A = s$.

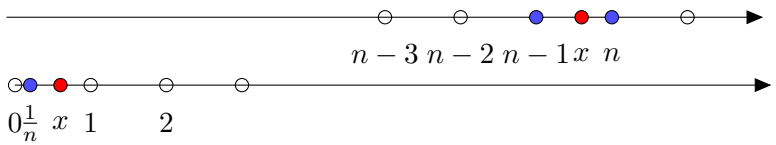


Figure 9: Any $x \in \mathbb{R}$ falls between $n-1$ and n (including equality) for some $n \in \mathbb{N}$. For any $x > 0$, there is $n \in \mathbb{N}$ such that $\frac{1}{n} < x$.

Theorem 20. *A real number that is has nonrepeating decimal representation is irrational.*

Proof. We prove that any (positive) rational number has a repeating decimal representation. Then the claim follows by contradiction.

Let $x = a_0.a_1a_2\cdots = \frac{p}{q}$, $p, q \in \mathbb{N}$. We can write $p = nq + r$, where $n, r \in \mathbb{N}$ and $0 \leq r < q$ (division with remainder). We set $a_0 = n$. Then we write $10r = n_1q + r_1$ again, and wet $a_1 = n_1$. In this way, we obtain the decimal representation of $\frac{p}{q}$, but there are only finitely many possible values $0, 1, \cdots q-1$ of r_1 because we are doing the division with remainder by q. This means that the numbers repeat after at largest q digits. □

The converse of this (any irrational number has a nonrepeating decimal representation) will be proven later.

The summation and product notations

Assume that we have a sequence of numbers, that is a family $\{a_n\}_{n\in S}$ of real numbers indexed by $S \subset \mathbb{N}$. This means that we have numbers $a_1, a_2, a_3, \cdots$. Sometimes we start the index from 0, and have $a_0, a_1, a_2, \cdots$.

Example 21.

- $a_1 = 1, a_2 = 2, a_3 = 3, \cdots$
- $a_1 = 1, a_2 = 4, a_3 = 9, \cdots$
- $a_1 = 4, a_2 = 2534, a_3 = \frac{3}{361}$ (a *finite* sequence stops at some $n \in \mathbb{N}$)

When we have a (finite) sequence, we can sum all these numbers up: $a_1 + \cdots + a_n$. We denote this by the following symbol.

$$\sum_{k=1}^{n} a_k = a_1 + \cdots + a_n$$

In this symbol, k is a dummy index and plays no specific role. We have

$$\sum_{k=1}^{n} a_k = \sum_{k=0}^{n-1} a_{k+1} = a_1 + \cdots + a_n.$$

On the other hand, the number on the top (n in this example) is where the sequence stops. Similarly, we can define $\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n$ for $n \geq m$.

More precisely, this is a recursive definition: We define $\sum_{k=1}^{1} = a_1$ and $\sum_{k=1}^{n+1} a_k = \sum_{k=1}^{n} a_k + a_{n+1}$ Similarly to mathematical induction, we define in this way $\sum_{k=1}^{n} a_k$ for all natural numbers $n \in \mathbb{N}$.

Example 22.

- $a_1 = 1, a_2 = 2, a_3 = 3$. $\sum_{k=1}^{3} a_k = 1 + 2 + 3 = 6$.
- $a_1 = 1, a_2 = 4, a_3 = 9, a_4 = 16$. $\sum_{k=1}^{4} a_k = 1 + 4 + 9 + 16 = 30$.

Let us also introduce a symbol for product.

$$\prod_{k=1}^{n} a_k = a_0 \cdot a_1 \cdot \cdots \cdot a_n$$

Example 23.

- $a_1 = 1, a_2 = 2, a_3 = 3$. $\prod_{k=1}^{3} a_k = 1 \cdot 2 \cdot 3 = 6$.
- $a_1 = 1, a_2 = 4, a_3 = 9, a_4 = 16$. $\prod_{k=1}^{4} a_k = 1 \cdot 4 \cdot 9 \cdot 16 = 576$.

In particular, we denote

- For $a \in \mathbb{R}$, $a^n = \prod_{k=1}^{n} a$. For example, $a^1 = a, a^2 = a \cdot a, a^3 = a \cdot a \cdot a$. By convention, for $a \neq 0$, we set $a^0 = 1$.
- $n! = \prod_{k=1}^{n} k = 1 \cdot 2 \cdot \cdots \cdot n$. By convention, we set $0! = 1$.

  For example, $2! = 2, 3! = 6, 4! = 24, \cdots$.

  $n!$ is the number of all possible ordering of n objects. One can see this by observing that in the first place there are n candidates, then in the second place there are $n-1$ remaining candidates, and so on.
- For $n, k \in \mathbb{N}, n \geq k$, we define $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. For example, $\binom{4}{2} = \frac{4!}{2!2!} = 6$.

  $\binom{n}{k}$ is the number of all possible combinations of choosing k objects out of n objects. One can see this by observing that, there are $n!$ orderings, and we can decided to pick the first k objects, but their order and the order of the remaining objects do not matter, thus we need to divide by $k!$ and $(n-k)!$.